Т6

$$p(x) = \begin{cases} \dfrac{\theta - 1}{x^{\theta}}, & x \geq 1 \\ 0, & x < 1 \end{cases}, \quad \theta > 1$$

$\vec{x}_n$ – выборка

$\Theta = (1, \infty)$ – открытое

$p(x)$ 2-ди непр. дифф. по θ

$\{x : p(x, \theta) > 0\}$ не зав. от θ

$$\frac{\partial}{\partial \theta} \int_{-\infty}^{\infty} p(x, \theta)\, dx = \frac{\partial}{\partial \theta} 1 = 0$$

$$\frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{\infty} p(x)\, dx = \frac{\partial^2}{\partial \theta^2} 1 = 0$$

$$\frac{\partial p}{\partial \theta} = \frac{(1 - \theta \ln x + \ln x)}{x^{\theta}}$$

$$\int_1^{\infty} \frac{\partial p}{\partial \theta}\, dx = \int_1^{\infty} \left[(1 - \theta) x^{-\theta} \ln x + x^{-\theta}\right] dx = \frac{x^{-\theta+1}}{1-\theta} \Bigg|_1^{+\infty} +$$

$$+ \int_1^{\infty} \ln x \, d(x^{1-\theta}) = \frac{1}{\theta - 1} - \int_1^{\infty} x^{-\theta}\, dx = 0$$

$$\frac{\partial^2 p}{\partial \theta^2} = \ln x \left(\frac{\theta \ln x - \ln x - 2}{x^{\theta}} \right)$$

$$\int_1^{\infty} \frac{\partial^2 p}{\partial x^2}\, dx = \int_1^{\infty} \left(\theta x^{-\theta} \ln^2 x - x^{-\theta} \ln^2 x - 2 x^{-\theta} \ln x\right) dx =$$

$$= \int_1^{\infty} \left((\theta - 1) x^{-\theta} \ln^2 x - 2 x^{-\theta} \ln x\right) dx = (\theta - 1) \frac{2}{(\theta - 1)^3} - 2 \frac{1}{(\theta - 1)^2} = 0$$

$\Rightarrow$ модель сильно регулярна

а) ОМП

$$L(\theta, \vec{x_i}) = \prod_{i=1}^{n} p(x_i, \theta) = \frac{(\theta-1)^n}{\left(\prod_{i=1}^{n} x_i\right)^{\theta}}$$

$$\ln L(\theta) = n \ln(\theta - 1) - \theta\left(\sum_{i=1}^{n} \ln x_i\right)$$

$$\frac{\partial \ln L}{\partial \theta} = \frac{n}{\theta - 1} - \sum_{i=1}^{n} \ln x_i = 0 \Rightarrow \theta = 1 + \frac{n}{\sum_{i=1}^{n} \ln x_i}$$

$$\frac{\partial^2 \ln L}{\partial \theta^2} = -\frac{n}{(\theta-1)^2} < 0 \Rightarrow \text{локальный max}$$

$$\tilde{\theta} = 1 + \frac{n}{\sum_{i=1}^{n} \ln x_i} = 1 + \frac{1}{\overline{\ln x}}$$

б) доверит. интервал для медианы

$$F(x) = \int_1^x \frac{\theta - 1}{t^{\theta}}\, dt = 1 - \frac{1}{x^{\theta-1}}, \quad x \geq 1$$

$$1 - \frac{1}{x_n^{\theta-1}} = \frac{1}{2} \Rightarrow x_n = 2^{\frac{1}{\theta-1}}$$

$$g(\tilde{\theta}) = 2^{\frac{1}{\tilde{\theta}-1}}$$

$$\sqrt{n}\, \frac{g(\tilde{\theta}) - g(\theta)}{\sqrt{\nabla^T g(\tilde{\theta}) I^{-1}(\tilde{\theta}) \nabla g(\tilde{\theta})}} \rightsquigarrow N(0,1)$$

$$I(\theta) = M\left[\left(\frac{\partial \ln p}{\partial \theta}\right)^2\right], \quad \ln p = \ln(\theta-1) - \theta \ln x$$

$$\left(\frac{\partial \ln p}{\partial \theta}\right)^2 = \left(\frac{1}{\theta-1} - \ln x\right)^2$$

$$I(\theta) = \int_1^\infty \left(\frac{1}{(\theta-1)^2} - 2\frac{\ln x}{\theta-1} + \ln^2 x\right)\frac{\theta-1}{x^\theta}\,dx =$$

$$= \int_1^\infty \left(\frac{x^{-\theta}}{\theta-1} - 2\frac{\ln x}{x^\theta} + (\theta-1)\frac{\ln^2 x}{x^\theta}\right)dx =$$

$$= \frac{1}{(\theta-1)^2} - 2\int_1^\infty \ln x\, d\left(-\frac{1}{(\theta-1)x^{\theta-1}}\right) + \int_1^\infty \frac{(\theta-1)\ln^2 x}{x^\theta}\,dx =$$

$$= \frac{1}{(\theta-1)^2} - 2\left[\ln \cancelto{0}{\frac{x^{1-\theta}}{1-\theta}}\Big|_1^\infty - \int_1^\infty \frac{x^{-\theta}}{1-\theta}\,dx\right] + \int_1^\infty \frac{(\theta-1)\ln^2 x}{x^\theta}\,dx =$$

$$= -\frac{1}{(1-\theta)^2} - \int_1^\infty \ln^2 x\, d(x^{1-\theta}) = -\frac{1}{(1-\theta)^2} - \left[\ln^2 x \cdot x^{1-\theta}\right]_1^\infty -$$

$$- 2\int_1^\infty \frac{\ln x}{x} x^{1-\theta}\,dx\Big] = -\frac{1}{(1-\theta)^2} + 2\int_1^\infty \ln x\, d\left(\frac{x^{1-\theta}}{1-\theta}\right) =$$

$$= -\frac{1}{(1-\theta)^2} + 2\left(\cancelto{0}{\frac{x^{1-\theta}\ln x}{1-\theta}}\Big|_1^\infty - \int_1^\infty \frac{x^{-\theta}}{1-\theta}\,dx\right) =$$

$$= -\frac{1}{(1-\theta)^2} + 2\frac{1}{(1-\theta)^2} = \frac{1}{(1-\theta)^2}$$ – непрер. на $(1, \infty)$

$\Rightarrow I(\theta)$ непрер. на $(1, \infty)$

$$\nabla g(\hat{\theta}) = 2^{\frac{1}{\hat{\theta}-1}} \cdot \ln(2) \cdot \frac{-1}{(\hat{\theta}-1)^2}$$

$$\sqrt{n}\,\frac{(g(\hat{\theta}) - g(\theta))(\hat{\theta}-1)}{2^{\frac{1}{\hat{\theta}-1}} \cdot \ln 2} \rightsquigarrow N(0,1)$$

$$g(\hat{\theta}) - \frac{1{,}96 \cdot 2^{\frac{1}{\hat{\theta}-1}} \cdot \ln 2}{\sqrt{n}\,(\hat{\theta}-1)} < x_m < g(\hat{\theta}) + \frac{1{,}96 \cdot 2^{\frac{1}{\hat{\theta}-1}} \ln 2}{\sqrt{n}\,(\hat{\theta}-1)}$$

c) $I(\theta) = \frac{1}{(1-\theta)^2}$

$$\frac{\sqrt{n}\,(\hat{\theta}-\theta)}{\hat{\theta}-1} \rightsquigarrow N(0,1)$$

$$t_1 < \sqrt{n}\,\overline{\ln x}\left(1 + \frac{1}{\overline{\ln x}} - \theta\right) < t_2$$

$$t_1 < \sqrt{n}\,((1-\theta)\,\overline{\ln x} + 1 < t_2$$

$$\frac{t_1}{\sqrt{n}} - 1 < (1-\theta)\,\overline{\ln x} < \frac{t_2}{\sqrt{n}} - 1$$

$$\frac{t_1 - \sqrt{n}}{\sqrt{n}\,\overline{\ln x}} < 1 - \theta < \frac{t_2 - \sqrt{n}}{\sqrt{n}\,\overline{\ln x}}$$

$$1 - \frac{t_2 - \sqrt{n}}{\sqrt{n}\,\overline{\ln x}} < \theta < 1 - \frac{t_1 - \sqrt{n}}{\sqrt{n}\,\overline{\ln x}}$$